SHGNet-56 Training Report

SHGNet-56 Training Report

Run name: `allsplits_46x_s123`
Generated: 2026-08-01 21:02
Project: 2shgnet-try-on (ear landmarks + piercing #56)
Status: Training **completed** (stages 1 → 2 → 3)

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1. Executive summary

This run fine-tuned a pretrained **SHGNet-56** hourglass on the full ear-pose dataset with **on-the-fly augmentation** (1 original + 45 structured variants = **46 samples per image per epoch**), using a **3-stage** unfreeze schedule.

Metric	Start (Stage 1 ep1)	Best Stage 1	Best Stage 2	Best Stage 3 (final)
Pierce error (px)	4.13	3.98	1.24	**1.14**
Landmark NME	0.0148	0.0143	0.0051	**0.0048**
PCK@0.05	0.979	0.983	0.997	**0.998**
Train loss	0.00065	—	—	0.00005

Headline result: piercing localization improved from ~**4.1 px** (Stage 1) to ~**1.14 px** (best Stage 3), with PCK@0.05 ≈ **0.998**.

Final weights: `outputs/checkpoints/SHGNet-56_final.pth` (from `best_stage3.pth`).

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2. Training command

```
cd "D:\try on proj\2shgnet-try-on"
.\.venv\Scripts\python.exe -u -m train.train `
  --device cuda `
  --variants-per-image 45 `
  --include-original `
  --all-splits `
  --train-on-all `
  --full-stages `
  --batch-size 16 `
  --run-name allsplits_46x_s123
```

Flag	Effect
`--device cuda`	Train on NVIDIA GPU (RTX 4090)
`--variants-per-image 45`	On-the-fly additive variants from `train/online_variants.py`
`--include-original`	Also train the unaugmented image → **46** samples/image
`--all-splits`	Pool `images/train` + `images/val` (+ `test` if present)

Flag	Effect
`--train-on-all`	No hold-out from training; val metrics on originals of the full set
`--full-stages`	Stage 1 (heads) → 2 (last hourglass) → 3 (full net) from SHGNet-56
`--batch-size 16`	Mini-batch size

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3. Dataset

Split folder	Annotated images
`data/data/ear_pose/images/train`	2,200
`data/data/ear_pose/images/val`	300
`test`	(none)
Total	**2,500**

Label format (YOLO-pose):
`class cx cy w h` + 56 × `(x y v)` normalized to [0, 1].
Piercing is landmark **#56** (0-based index **55**).

Per-epoch sample count:

Item	Count
Base images	2,500
Samples / image	46 (1 original + 45 augs)
Train set length	**115,000**
Steps / epoch (bs=16)	**7,188**
Val set (originals only)	2,500

Online augmentation families (45 additive variants)

Family	Count	Notes
Flip	2	incl. identity / horizontal
Rotation	7	≈ ±15°
Scale	5	≈ 0.85–1.15

Family	Count	Notes
Translate	9	small shifts
Brightness	5	
Contrast	5	
Blur	3	kernels 3/5/7
Noise / JPEG	4	
Occlusion (hair-like)	5	
Total	**45**	+ original = **46**

Variants are applied **in RAM** (no disk dump). Validation uses **originals only**.

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4. Model and stages

Architecture: LDNet / stacked hourglass, **2 stacks**, input **256×256**, heatmaps **64×64**, **56** landmark channels.

Init checkpoint: `models/shgnet/SHGNet-56_final.pth` (pretrained before this run).

Stage	What trains	Epochs	LR	Trainable params (this run)
1	Heatmap output heads only	30	1e-3	28,784
2	Last hourglass + features + heads	20	1e-4	4,019,568
3	Full network	15	1e-5	8,459,888

Loss: heatmap MSE (Gaussian targets, σ=2).
Best checkpoint selection: lowest `piercing_point_error_px` on the val originals set.
No early stopping / patience — fixed epoch counts.

Hardware: NVIDIA GeForce RTX 4090 (~24 GB), CUDA; observed ~4.5–5.7 it/s; ~26–30 min per epoch (train + val). Peak VRAM use was modest (~few GB) because the model is small and Stage 1 only trains heads.

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5. Best metrics by stage

Stage	Best ep	Train loss	NME	Pierce px	PCK@0.05
stage1	27	0.000619	0.0143	3.979	0.9827
stage2	19	0.000074	0.0051	1.242	0.9974
stage3	14	0.000046	0.0048	1.144	0.9975

Improvement vs Stage 1 start

- Pierce px: 4.13 → **1.14** (~72.3% relative reduction)
- NME: 0.0148 → **0.0048**

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6. Per-epoch logs

Stage 1 (heads)

Ep	Train loss	Val HM loss	NME	Pierce px	PCK@0.05
1	0.000650	0.000601	0.0148	4.130	0.9790
2	0.000639	0.000599	0.0147	4.179	0.9797
3	0.000636	0.000591	0.0146	4.130	0.9802
4	0.000633	0.000589	0.0146	4.057	0.9803
5	0.000631	0.000590	0.0145	4.104	0.9812
6	0.000630	0.000588	0.0145	3.986	0.9811
7	0.000629	0.000586	0.0145	4.079	0.9814
8	0.000628	0.000583	0.0145	4.080	0.9813
9	0.000627	0.000581	0.0144	4.006	0.9816
10	0.000626	0.000582	0.0144	4.110	0.9818
11	0.000626	0.000580	0.0144	4.002	0.9818
12	0.000625	0.000605	0.0144	4.099	0.9816
13	0.000624	0.000578	0.0144	4.091	0.9815
14	0.000624	0.000581	0.0144	4.050	0.9818
15	0.000623	0.000579	0.0144	4.074	0.9820
16	0.000623	0.000577	0.0143	4.087	0.9821

Ep	Train loss	Val HM loss	NME	Pierce px	PCK@0.05
17	0.000623	0.000583	0.0143	4.119	0.9821
18	0.000622	0.000578	0.0143	4.078	0.9824
19	0.000622	0.000579	0.0143	4.108	0.9824
20	0.000621	0.000583	0.0143	4.058	0.9824
21	0.000621	0.000578	0.0143	4.068	0.9822
22	0.000621	0.000575	0.0143	3.991	0.9826
23	0.000621	0.000576	0.0143	4.039	0.9826
24	0.000621	0.000576	0.0143	4.039	0.9827
25	0.000620	0.000573	0.0142	4.101	0.9826
26	0.000620	0.000571	0.0143	4.059	0.9829
27	0.000619	0.000576	0.0143	3.979	0.9827
28	0.000619	0.000571	0.0142	4.015	0.9828
29	0.000619	0.000575	0.0143	4.140	0.9827
30	0.000619	0.000579	0.0143	4.122	0.9825

Stage 2 (last hourglass)

Ep	Train loss	Val HM loss	NME	Pierce px	PCK@0.05
1	0.000441	0.000265	0.0097	2.939	0.9934
2	0.000280	0.000158	0.0077	2.536	0.9951
3	0.000214	0.000124	0.0069	2.235	0.9959
4	0.000179	0.000102	0.0065	1.996	0.9964
5	0.000156	0.000084	0.0061	1.765	0.9968
6	0.000139	0.000076	0.0060	1.613	0.9969
7	0.000128	0.000066	0.0056	1.524	0.9971
8	0.000118	0.000064	0.0057	1.517	0.9970
9	0.000110	0.000057	0.0055	1.432	0.9972
10	0.000104	0.000055	0.0055	1.389	0.9972
11	0.000099	0.000055	0.0055	1.454	0.9974
12	0.000094	0.000049	0.0054	1.341	0.9972
13	0.000090	0.000046	0.0053	1.325	0.9973
14	0.000087	0.000044	0.0052	1.318	0.9974

Ep	Train loss	Val HM loss	NME	Pierce px	PCK@0.05
15	0.000084	0.000044	0.0053	1.280	0.9974
16	0.000081	0.000042	0.0052	1.272	0.9975
17	0.000079	0.000041	0.0053	1.295	0.9974
18	0.000076	0.000038	0.0052	1.257	0.9974
19	0.000074	0.000037	0.0051	1.242	0.9974
20	0.000072	0.000037	0.0049	1.243	0.9979

Stage 3 (full network)

Ep	Train loss	Val HM loss	NME	Pierce px	PCK@0.05
1	0.000060	0.000025	0.0050	1.188	0.9974
2	0.000056	0.000025	0.0049	1.177	0.9975
3	0.000054	0.000024	0.0049	1.165	0.9974
4	0.000053	0.000023	0.0049	1.161	0.9974
5	0.000052	0.000023	0.0049	1.153	0.9974
6	0.000051	0.000022	0.0049	1.150	0.9974
7	0.000050	0.000022	0.0048	1.155	0.9975
8	0.000050	0.000022	0.0049	1.150	0.9974
9	0.000049	0.000022	0.0049	1.148	0.9975
10	0.000048	0.000021	0.0049	1.152	0.9975
11	0.000048	0.000021	0.0047	1.149	0.9977
12	0.000047	0.000021	0.0048	1.149	0.9974
13	0.000047	0.000021	0.0048	1.150	0.9974
14	0.000046	0.000020	0.0048	1.144	0.9975
15	0.000046	0.000021	0.0049	1.153	0.9974

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7. Artifacts

File	Role	Size (MB)
`outputs/checkpoints/best_stage1.pth`	checkpoint	32.7
`outputs/checkpoints/best_stage2.pth`	checkpoint	32.7

File	Role	Size (MB)
outputs/checkpoints/best_stage3.pth	checkpoint	32.7
outputs/checkpoints/SHGNet-56_final.pth	checkpoint	32.7
outputs/checkpoints/last_stage1.pth	checkpoint	32.7
outputs/checkpoints/last_stage2.pth	checkpoint	32.7
outputs/checkpoints/last_stage3.pth	checkpoint	32.7
outputs/train_results.json	full history + best dicts	—
models/shgnet/SHGNet-56_final.pth	(pre-run init; may be overwritten if re-wired)	—

Recommended for export / live:

```
.\.venv\Scripts\python.exe -m train.export_onnx `
  --checkpoint outputs/checkpoints/SHGNet-56_final.pth `
  --out models/shgnet/SHGNet-56.onnx
```

Then:

```
.\.venv\Scripts\python.exe -m live.desktop_onnx
```

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8. Pipeline diagram

```
Ear crops + YOLO labels (#56)
        |
        ▼
Pool train+val (2,500) ──▶ Piercing56Dataset
        |                     1 original + 45 online variants
        |                     → 115,000 samples / epoch
        ▼
Stage 1 (30 ep, lr 1e-3)  heads only
        ▼
Stage 2 (20 ep, lr 1e-4)  last hourglass + heads
        ▼
Stage 3 (15 ep, lr 1e-5)  full network
```

▼

```
best_stage3.pth → SHGNet-56_final.pth → ONNX → live / web try-on
```

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9. Code entry points

Topic	Path
Train CLI	`train/train.py`
Dataset + 46× expansion	`train/dataset.py`
Online variants	`train/online_variants.py`
Config (epochs, LR, paths)	`train/config.py`
Freeze / unfreeze stages	`train/model.py`
Workflow notes	`WORKFLOW.md`
Results JSON	`outputs/train_results.json`

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10. Notes and caveats

1. **`--train-on-all`:** validation metrics are computed on originals of images that also appear in training (with augs). Absolute numbers are optimistic vs a strict held-out test set; relative improvement across stages remains meaningful.
2. **Stage 1 looked flat** because the network was already strong and only heads were trainable; **Stage 2** delivered the large pierce-error drop.
3. **No patience / early stop** — all 65 epochs ran to completion.
4. Re-running with a larger `--batch-size` (e.g. 32–64) may improve throughput on the 4090; this run used 16.

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11. Conclusion

The `allsplits_46x_s123` run successfully completed 3-stage fine-tuning of SHGNet-56 on 2,500 ear images with 46× on-the-fly sampling. Piercing error improved to **~1.14 px** (best Stage 3) with **PCK@0.05 ≈ 0.998**. Artifacts are ready for ONNX export and live try-on.

Report auto-generated from `outputs/train_results.json` by `scripts/build_training_report.py`.